

DEMOGRAPHIC INFORMATION

PATIENT

NAME	[REDACTED]
DATE OF BIRTH	[REDACTED]
SEX	[REDACTED]
MEDICAL RECORD #	M003487582
ACCESSION #	[REDACTED]
LAB NUMBER	[REDACTED]
FAMILY NUMBER	[REDACTED]

TEST INFORMATION

TEST NAME	Rapid Trio WGS
TEST CODE	1822
SAMPLE TYPE	[REDACTED]
DATE COLLECTED	08/20/2025
DATE RECEIVED	08/28/2025
DATE REPORTED	09/02/2025

RECIPIENT

PHYSICIAN NAME	David Dimmock
FACILITY	VCHC
LOCATION	
PHONE	- -
FAX	559-353-6087

ADDITIONAL RECIPIENT

NAME	LIANN JIMMONS
PHONE	
FAX	559-353-6087

ADDITIONAL RECIPIENT

NAME	Jason Carmichael
PHONE	
FAX	559-353-6087

CLINICAL INDICATION

Based on the submitted clinical information, the patient has hypotonia, poor suck, failure to thrive, laryngomalacia, respiratory failure, feeding difficulties, sacral dimple, stridor and tracheal tug on inspiration.

We have also received a sample from the father (DNA#3455730) and the mother (DNA#3455728) of this individual.

RESULTS



DISEASE	INHERITANCE PATTERN	GENE/ VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
15q11.2q13.1 Deletion Syndrome	Autosomal Dominant	15q11.2q13.1:HG3 8 chr15:22650252-28293509 DEL 5.643(Mb)	Copy Number Variant	Heterozygous	De Novo	Pathogenic

RESULTS SUMMARY

A heterozygous pathogenic deletion within chromosome 15q11.2q13.1 was detected. The genome data indicate this deletion arose on the paternal allele, consistent with a diagnosis of Prader-Willi syndrome.

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ADDITIONAL FINDINGS

ACMG recommended secondary findings (v3.2; PMID 37347242)

Client has not selected to receive these findings.

Other Incidental Findings

Client has not selected to receive these findings.

Research Findings

Client has not selected to receive these findings.

RECOMMENDATIONS

- It is recommended that correlation of these findings with the clinical phenotype be performed.
- Genetic counseling is recommended.
- For sequence variants, testing is available for this individual's at risk relatives under test code 1580. Visit <https://www.baylorgenetics.com/known-familial-mutations-policy/> for further information. Please consult with the lab for available test codes for other types of variants (CNV, STR, mito).
- Contact laboratory for requests regarding reanalysis and raw data.

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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE/ VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
15q11.2q13.1 Deletion Syndrome	Autosomal Dominant	15q11.2q13.1:HG38 chr15:22650252-28293509 DEL 5.643(Mb)	Copy Number Variant	Heterozygous	De Novo	Pathogenic

GENE DESCRIPTION

This interval is located between chromosome bands 15q11.2 and 15q13.1, spanning approximately 5.643 Mb in size. This is a recurrent deletion mapped to chromosome 15q11q13 type I deletion between low copy repeats BP1 and BP3, which is a well-known imprinted region and includes the SNRPN and UBE3A genes among many other protein-coding genes.

DISEASE DESCRIPTION

Deletions in this region are consistent with the clinical diagnosis of Prader-Willi syndrome (PWS) or Angelman syndrome (AS). Deletion of the paternally-inherited allele may cause PWS (OMIM: 176270), which is characterized by weak cry, severe hypotonia, and feeding difficulties in infancy, uncontrolled hyperphagia leading to obesity in early childhood, short stature, global developmental delay, mild intellectual disability, small hands and feet, hypogonadism associated with infertility, growth hormone deficiency, and behavioral abnormalities. Facial dysmorphism may include narrow forehead, almond-shaped palpebral fissures, narrow nasal bridge, and thin vermilion of the upper lip with down-turned corners of the mouth. Deletion of the maternally-inherited allele may cause AS (OMIM: 105830), which is characterized by microcephaly, developmental delay, severe intellectual disability, severe speech impairment, ataxia, seizure, and unique behavior with an apparent happy demeanor that includes frequent laughing, smiling, and excitability.

VARIANT DETAILS - 15q11.2q13.1:HG38 chr15:22650252-28293509 DEL 5.643(Mb)

- (GRCh38)chr15:22650252-28293509DEL -
- This deletion is consistent with the recurrent BP1-BP3 deletion in the PWS/AS critical region on chromosome 15, mediated by flanking low copy repeats.
- The genome data indicate this deletion occurred on the paternal allele, consistent with a diagnosis of PWS.

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QUALITY METRICS

Percentage depth of coverage at 20x: **99.64%**
Mean depth of coverage: **55.34x**

METHODOLOGY

Genomic DNA was fragmented and indexed to prepare individual libraries. The libraries were purified and pooled for sequence analysis on the Illumina NovaSeq platform for paired end reads. Internal quality control measures are performed to ensure sample integrity.

Data analysis and interpretation are performed by the Baylor Genetics analytics pipeline. Data are aligned to the human reference genome build GRCh38 using the Illumina Dragen BioIT Platform. Variant calling is performed using the Illumina Dragen haplotype-based variant calling system and Illumina Dragen genome wide depth based CNV (Copy Number Variants) caller with custom modifications from Baylor Genetics. Short tandem repeat calling is performed using the Illumina Expansion Hunter tool.

The automated genetic interpretation platform from Emedgene is implemented for variant annotation and interpretation with proprietary algorithms and open-source data sets such as gnomAD, EVS (Exome Variant Server), ClinVar, and professional resources such as HGMD (Human Gene Mutation Database) Pro. The variants were interpreted according to ACMG (American College of Medical Genetics) guidelines (reference 1) and patient phenotypes.

Variants are confirmed by Sanger sequencing, specific PCR, Multiplex Ligation-dependent Probe Amplification (MLPA), or another methodology, if sequencing or copy number variants detected by NGS do not meet internal quality standards or are within highly homologous regions.

Synonymous variants, noncoding variants not affecting splicing sites, and common benign variants are excluded from interpretation unless previously reported in the literature as possibly pathogenic variants. This test may not provide detection of certain variants or portions of certain genes due to insufficient sequencing depth, local sequence characteristics, high/low genomic complexity, or the presence of closely related pseudogenes. Small deletions or insertions larger than 25bp, low-level mosaic variants, structural variants such as inversions, and/or balanced translocations may not be detected with this technology.

Variant types being analyzed include sequencing variants and CNV in nuclear genes and mitochondrial DNA (mtDNA), and short tandem repeats (STR). Uniparental disomy (UPD) analysis is done only in a trio setting. Regions of homozygosity (ROH) are reported when greater than 5 Mb. Sensitivity of detection is decreased for mtDNA sequencing variants and CNVs at heteroplasmy levels lower than 5% and 10%, respectively. Pathogenic events of short tandem repeats can be detected within the genomic regions of following genes: AFF2, AR, ATN1, ATXN1, ATXN2, ATXN3, ATXN7, ATXN10, ATXN8OS, C9orf72, CACNA1A, CNBP, CSTB, DIP2B, DMPK, FGF14, FMR1, FXN, GLS, HTT, JPH3, NOP56, NOTCH2NLC, PABPN1, PHOX2B, PPP2R2B, RFC1, TBP, and TCF4. Sensitivity of detection is reduced for borderline or incomplete penetrance alleles.

The proband report contains results of genes related to the patient's clinical phenotype. Variants relevant to the phenotype are reported as positive (pathogenic/likely pathogenic) or variants of uncertain significance (VUS). Benign and likely benign variants are not reported. This interpretation is based on the clinical information provided by the referring institution and our current understanding of the gene and variants at the time of reporting. Additional methodologies may be used to aid in variant interpretation. Results in the report should be carefully correlated to the individual's clinical findings by the referring physician.

If elected, secondary findings including pathogenic and likely pathogenic variants in genes included in the ACMG secondary findings recommendations, or other findings determined to be severe and medically significant (incidental findings), will be reported. The ACMG recommendations (ACMG SF v3.2 (reference 2)) include the following genes: ACTA2, ACTC1, ACVRL1, APC, APOB, ATP7B, BAG3, BMPR1A, BRCA1, BRCA2, BTBD9, CACNA1S, CALM1, CALM2, CALM3, CASQ2, COL3A1, DES, DSC2, DSG2, DSP, ENG, FBN1, FLNC, GAA, GLA, HFE, HNF1A, KCNH2, KCNQ1, LDLR, LMNA, MAX, MEN1, MLH1, MSH2, MSH6, MUTYH, MYBPC3, MYH11, MYH7, MYL2, MYL3, NF2, OTC, PALB2, PCSK9, PKP2, PMS2, PRKAG2, PTEN, RB1, RBM20, RET, RPE65, RYR1, RYR2, SCN5A, SDHAF2, SDHB, SDHC, SDHD, SMAD3, SMAD4, STK11, TGFBF1, TGFBF2, TMEM127, TMEM43, TNNC1, TNNI3, TNNT2, TP53, TPM1, TRDN, TSC1, TSC2, TTN, TTR, VHL, and WT1. Variants unrelated to the patient's phenotype but potentially clinically significant in genes with no known disease association will be reported if elected.

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LIMITATIONS

The interpretation of nucleotide changes is based on our current understanding of the specific genes. This interpretation may change over time as more information about these genes becomes available. Possible diagnostic errors include sample mix-ups, genetic variants that interfere with analysis, incorrect assignment of biological parentage, or other sources. Please contact a genetic counselor at Baylor Genetics if there is reason to suspect one of these sources of error.

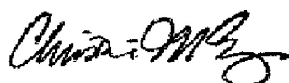
REFERENCES

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015 May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.
2. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, Klein TE, Richards CS, Stewart DR, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2023 Aug;25(8):100866. doi: 10.1016/j.gim.2023.100866. PMID: 37347242.

DISCLAIMER

This test was developed, and its performance was determined by the Baylor Miraca Genetics Laboratory DBA Baylor Genetics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). Since FDA is not required for clinical use of this test, this laboratory has established and validated the test's accuracy and precision, pursuant to the requirement of CLIA'88. This laboratory is licensed and/or accredited under CLIA and CAP (CAP#2109314 / CLIA# 45D0660090; Lab Director: Christine M. Eng, MD).

ELECTRONICALLY SIGNED BY



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